

Lecture 14

Sliding Mode Observer

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E1 248
Sliding mode control and its application

- 1 Sliding Mode Observer
- 2 Sliding Mode Observers for Linear Uncertain Systems

Sliding Mode Observer: Utkin's Observer (1981)

Consider a linear time-invariant system

$$\dot{x} = Ax + Bu$$

$$y = Cx$$

where $u \in \mathbb{R}^m$, $y \in \mathbb{R}^p$ and $p \geq m$.

Sliding mode observer is structurally similar except in their innovative correction term. At first, the system is transformed to

$$\dot{x}_1 = A_{11}x_1 + A_{12}y + B_1u$$

$$\dot{y} = A_{21}x_1 + A_{22}y + B_2u.$$

Remark

There always exist a transformation such that $x \mapsto Tx =: (x_1, y)$ and this transformation is given as

$$T = \begin{bmatrix} N_c^\top \\ C \end{bmatrix}$$

where $N_c \in \mathbb{R}^{n \times (n-p)}$ spans the null space of C . The new output matrix is $CT^{-1} = [0 \quad I_p]$.

Sliding Mode Observer

The sliding mode observer is proposed as

$$\begin{aligned}\dot{\hat{x}}_1 &= A_{11}\hat{x}_1 + A_{12}\hat{y} + B_1u - L_1L_2\text{sign}(y - \hat{y}) \\ \dot{\hat{y}} &= A_{21}\hat{x}_1 + A_{22}\hat{y} + B_2u + L_2\text{sign}(y - \hat{y})\end{aligned}$$

where gain L_2 is chosen such that sliding mode occurs in finite time on the manifold

$$S = \{(e_1, e_y) : e_y = 0\}.$$

Here $e_1 = x_1 - \hat{x}_1$ and $e_y = y - \hat{y}$.

Observer Error Dynamics

Using the error variables, the error dynamics can be given as

$$\begin{aligned}\dot{e}_1 &= A_{11}e_1 + A_{12}e_y + L_1L_2\text{sign}(e_y) \\ \dot{e}_y &= A_{21}e_1 + A_{22}e_y - L_2\text{sign}(e_y).\end{aligned}$$

Now selecting the gain L_2 such that $L_2 > A_{21}e_1 + A_{22}e_y$, we obtain the error dynamics as

$$\dot{e}_y \leq -\eta\text{sign}(e_y).$$

Sliding Mode Observer

Observer Error Dynamics

Thus, $e_y^\top \dot{e}_y \leq -\eta \|e_y\|$ and achieve $\hat{y} = y$ in finite time. When $e_y = 0$, equivalent control is

$$L_2 \text{sign}(e_y)|_{\text{eq}} = A_{21} e_1.$$

Then, reduced dynamics is given as

$$\dot{e}_1 = A_{11} e_1 + L_1 A_{21} e_1 = (A_{11} + L_1 A_{21}) e_1.$$

Design L_1 such that $A_{11} + L_1 A_{21}$ is Hurwitz and e_1 goes to zero asymptotically.

Remark

It is to be noted that for the sliding mode to occur, i.e., $e_y = 0$, the $(e_1, e_y) \in \Omega$, where

$$\Omega = \{(e_1, e_y) : \|A_{21}\| \|e_1\| + \lambda_{\max}\{A_{22}\} \|e_y\| < L_2 - \eta\}.$$

for any $\eta < L_2$.

Sliding Mode Observer

Another Approach

- Consider the observer error dynamics

$$\begin{aligned}\dot{e}_1 &= A_{11}e_1 + A_{12}e_y + L_1L_2\text{sign}(e_y) \\ \dot{e}_y &= A_{21}e_1 + A_{22}e_y - L_2\text{sign}(e_y).\end{aligned}$$

- Define transformation $(e_1, e_y) \mapsto T(e_1, e_y) =: (\tilde{e}_1, \tilde{e}_y) = (\tilde{e}_1, e_y)$ where

$$T = \begin{bmatrix} I_{n-p} & L_1 \\ 0 & I_p \end{bmatrix}.$$

- Using this transformation, we obtain

$$\begin{aligned}\dot{\tilde{e}}_1 &= \tilde{A}_{11}\tilde{e}_1 + \tilde{A}_{12}e_y \\ \dot{e}_y &= A_{21}\tilde{e}_1 + \tilde{A}_{22}e_y - L_2\text{sign}(e_y).\end{aligned}$$

where $\tilde{A}_{11} = A_{11} + L_1A_{21}$, $\tilde{A}_{12} = \tilde{A}_{11}L_1 + A_{12} + L_1A_{22}$ and $\tilde{A}_{22} = A_{21}L_1 + A_{22}$. Define the set

$$\tilde{\Omega} = \{(\tilde{e}_1, e_y) : \|A_{21}\|\|\tilde{e}_1\| + \|\tilde{A}_{22}\|\|e_y\| < L_2 - \eta\}.$$

Sliding Mode Observer

- Now design $L_2 > A_{21}\tilde{e}_1 + \tilde{A}_{22}e_y$ such that $(\tilde{e}_1, e_y) \in \tilde{\Omega}$. Then,

$$\dot{e}_y \leq -\eta \text{sign}(e_y)$$

and finally $e_y^\top \dot{e}_y \leq -\eta \|e_y\|$. Thus, $e_y = 0$ in finite time.

- The reduced order dynamics is given as

$$\dot{\tilde{e}}_1 = \tilde{A}_{11}\tilde{e}_1.$$

- Since (A, C) is observable pair, it implies that (A_{11}, A_{21}) is also observable pair. Then, L_1 can be chosen such that $\tilde{A}_{11}$ is Hurwitz and $\tilde{e}_1$ goes to zero asymptotically.
- Eventually, it makes e_1 to go to zero asymptotically.¹²

¹S. Drakunov and V. Utkin, "Sliding mode observer: tutorial", *In the Proc. 34th Conf. Dec. Control*, New Orleans, LA, USA, Dec, 1995.

²[Chap. 6]C. Edwards and S. K. Spurgeon, *Sliding Mode Control: Theory and Application*, CRC Press, Taylor and Francis Group, New York, 1998.

Sliding Mode Observer

Remarks

- In the following dynamics

$$\ddot{\tilde{e}}_1 = \tilde{A}_{11}\tilde{e}_1 + \tilde{A}_{12}e_y$$

$$\dot{e}_y = A_{21}\tilde{e}_1 + \tilde{A}_{22}e_y - L_2\text{sign}(e_y)$$

the variable $\tilde{e}_1$ converges to zero asymptotically only after the sliding mode is enforced, i.e., $e_y = 0$. So, the trajectory $\tilde{e}_1$ may not decrease until the sliding mode starts.

- A modification to this observer is proposed by Edwards and Spurgeon^a to overcome this problem. It is shown that with this modified observer, the estimation error converges to zero even before sliding mode takes place.

^aC. Edwards and S. Spurgeon, "On the development of discontinuous observers", *International Journal of Control*, vol. 59, no. 5, pp. 1211–1229, 1994.

Sliding Mode Observer

Modified Observer

The modified observer is given with linear injection terms as

$$\begin{aligned}\dot{\tilde{e}}_1 &= \tilde{A}_{11}\tilde{e}_1 + \tilde{A}_{12}e_y - G_1e_y \\ \dot{e}_y &= A_{21}\tilde{e}_1 + \tilde{A}_{22}e_y - G_2e_y - L_2\text{sign}(e_y).\end{aligned}$$

Here, the gains are selected as $G_1 = \tilde{A}_{12}$ and $G_2 = \tilde{A}_{22} - A_{22}^s$, where A_{22}^s is any stable design matrix of appropriate dimension. Using this, the above dynamics reduces to

$$\begin{aligned}\dot{\tilde{e}}_1 &= \tilde{A}_{11}\tilde{e}_1 \\ \dot{e}_y &= A_{21}\tilde{e}_1 + A_{22}^s e_y - L_2\text{sign}(e_y).\end{aligned}$$

- The error system is asymptotically stable for $L_2 \equiv 0$ because the poles of the combined system are given by $\lambda\{\tilde{A}_{11}\} \cup \lambda\{A_{22}^s\}$ and so lie in the open left half complex plane.
- In the original Utkin observer, the switching action was potentially required to make the error system stable.
- The addition of a Luenberger type gain matrix, feeding back the output error, yields the potential to provide robustness against certain classes of uncertainty by virtue of the discontinuous component $L_2\text{sign}(e_y)$.

Sliding Mode Observers for Linear Uncertain Systems

Synthesis of a Discontinuous Observer

Consider the dynamical system

$$\begin{aligned}\dot{x}(t) &= Ax(t) + Bu(t) + D\xi(t, x, u) \\ y(t) &= Cx(t)\end{aligned}$$

where $A \in \mathbb{R}^{n \times n}$, $B \in \mathbb{R}^{n \times m}$, $C \in \mathbb{R}^{p \times n}$ and $D \in \mathbb{R}^{n \times q}$ where $p \geq q$. The function $\xi : \mathbb{R}_+ \times \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^q$ is unknown but bounded so that

$$\|\xi(t, x, u)\| \leq r_1 \|u\| + \alpha(t, y)$$

where r_1 is a known scalar and $\alpha : \mathbb{R}_+ \times \mathbb{R}^p \rightarrow \mathbb{R}_+$ is a known function.

The system can be written as

$$\begin{aligned}\dot{x}_1(t) &= A_{11}x_1(t) + A_{12}y(t) + B_1u(t) \\ \dot{y}(t) &= A_{21}x_1(t) + A_{22}y(t) + B_2u(t) + D_2\xi.\end{aligned}$$

where $x_1 \in \mathbb{R}^{(n-p)}$, $y \in \mathbb{R}^p$ and the matrix A_{11} has stable eigenvalues.

Sliding Mode Observers for Linear Uncertain Systems

A canonical Form for Observer Design

Consider the observer of the form

$$\begin{aligned}\dot{\hat{x}}_1(t) &= A_{11}\hat{x}_1(t) + A_{12}\hat{y}(t) + B_1u(t) - A_{12}e_y(t) \\ \dot{\hat{y}}(t) &= A_{21}\hat{x}_1(t) + A_{22}\hat{y}(t) + B_2u(t) - (A_{22} - A_{22}^s)e_y(t) + \nu\end{aligned}$$

where A_{22}^s is a stable design matrix and $e_y = \hat{y} - y$. Let $P_2 \in \mathbb{R}^{p \times p}$ be symmetric positive definite Lyapunov matrix for A_{22}^s .

Define discontinuous vector ν as

$$\nu = \begin{cases} -\rho(t, y, u) \|D_2\| \frac{P_2 e_y}{\|P_2 e_y\|}, & \text{if } e_y \neq 0 \\ 0, & \text{otherwise} \end{cases}$$

where the scalar function $\rho : \mathbb{R}_+ \times \mathbb{R}^p \times \mathbb{R}^m \rightarrow \mathbb{R}_+$ satisfies

$$\rho(t, x, u) \geq r_1 \|u\| + \alpha(t, y) + \gamma_0$$

γ_0 is a positive scalar.

Sliding Mode Observers for Linear Uncertain Systems

If state estimation error $e_1 = \hat{x}_1 - x_1$, then

$$\dot{e}_1(t) = A_{11}e_1(t)$$

$$\dot{e}_y(t) = A_{21}e_1(t) + A_{22}^s e_y(t) + \nu - D_2\xi$$

Proposition

There exists a family of symmetric positive definite matrices P_2 such that the uncertain dynamical error system above is quadratically stable.

Proof

- Let $Q_1 \in \mathbb{R}^{(n-p) \times (n-p)}$ and $Q_2 \in \mathbb{R}^{p \times p}$ be symmetric positive definite design matrices.
- Define $P_2 \in \mathbb{R}^{p \times p}$ to be the unique symmetric positive definite solution to the Lyapunov equation

$$P_2 A_{22}^s + (A_{22}^s)^T P_2 = -Q_2$$

Sliding Mode Observers for Linear Uncertain Systems

Proof

Using the computed value of P_2 define

$$\hat{Q} = A_{21}^T P_2 Q_2^{-1} P_2 A_{21} + Q_1$$

and $\hat{Q} = \hat{Q}^T > 0$.

Let $P_1 \in \mathbb{R}^{(n-p) \times (n-p)}$ be the unique symmetric positive definite solution to the Lyapunov equation

$$P_1 A_{11} + A_{11}^T P_1 = -\hat{Q}$$

Consider the candidate Lyapunov function

$$V(e_1, e_y) = e_1^T P_1 e_1 + e_y^T P_2 e_y$$

Taking time derivative of the Lyapunov function along the system trajectory

$$\dot{V} = -e_1^T \hat{Q} e_1 + e_1^T A_{21}^T P_2 e_y + e_y^T P_2 A_{21} e_1 - e_y^T Q_2 e_y + 2e_y^T P_2 \nu - 2e_y^T P_2 D_2 \xi$$

Sliding Mode Observers for Linear Uncertain Systems

Proof

It is easy to verify that

$$\begin{aligned} (e_y - Q_2^{-1} P_2 A_{21} e_1)^T Q_2 (e_y - Q_2^{-1} P_2 A_{21} e_1) \\ = e_y^T Q_2 e_y - e_1^T A_{21}^T P_2 e_y - e_y^T P_2 A_{21} e_1 + e_1^T A_{21}^T P_2 Q_2^{-1} P_2 A_{21} e_1. \end{aligned}$$

Denote $\tilde{e}_y = e_y - Q_2^{-1} P_2 A_{21} e_1$. Then,

$$\begin{aligned} \dot{V} &= -e_1^T \hat{Q} e_1 + e_1^T A_{21}^T P_2 Q_2^{-1} P_2 A_{21} e_1 - \tilde{e}_y^T Q_2 \tilde{e}_y + 2e_y^T P_2 \nu - 2e_y^T P_2 D_2 \xi \\ &= -e_1^T Q_1 e_1 - \tilde{e}_y^T Q_2 \tilde{e}_y + 2e_y^T P_2 \nu - 2e_y^T P_2 D_2 \xi \\ &= -e_1^T Q_1 e_1 - \tilde{e}_y^T Q_2 \tilde{e}_y - 2\rho(t, x, u) \|D_2\| \|P_2 e_y\| - 2e_y^T P_2 D_2 \xi. \end{aligned}$$

In the last equality, we use $e_y^T P P e_y = (e_y^T P)(e_y^T P)^T = \|e_y^T P\|^2$.

Recalling the fact that $\rho(t, x, u) \geq r_1 \|u\| + \alpha(t, y) + \gamma_0$, above equation becomes

$$\begin{aligned} \dot{V} &\leq -e_1^T Q_1 e_1 - \tilde{e}_y^T Q_2 \tilde{e}_y - 2\rho(t, x, u) \|D_2\| \|P_2 e_y\| + 2\|D_2\| (r_1 \|u\| + \alpha(y)) \|P_2 e_y\| \\ &\leq -e_1^T Q_1 e_1 - \tilde{e}_y^T Q_2 \tilde{e}_y - 2\gamma_0 \|D_2\| \|P_2 e_y\| < 0. \end{aligned}$$

Hence, the error system is quadratically stable.

Sliding Mode Observers for Linear Uncertain Systems

Sliding Motion

Consider the hyperplane in the error space given by

$$S_0 = \{e \in \mathbb{R}^n : Ce = 0\}$$

Corollary

An ideal sliding motion takes place on S_0 with the above designed ν .

Proof

Consider the Lyapunov function $V_s(e_y) = e_y^T P_2 e_y$.

Differentiating

$$\begin{aligned} \dot{V}_s &= -e_y^T Q_2 e_y + 2e_y^T P_2 A_{21} e_1 + 2e_y^T P_2 (\nu - D_2 \xi) \\ &\leq 2 \|P_2 e_y\| \|A_{21} e_1\| - 2\gamma_0 \|D_2\| \|P_2 e_y\|. \end{aligned}$$

Sliding Mode Observers for Linear Uncertain Systems

Proof

Now, we see that

$$\|P_2 e_y\|^2 = \left(P_2^{1/2} e_y\right)^T P_2 \left(P_2^{1/2} e_y\right) \geq \lambda_{\min}(P_2) \left\|P_2^{1/2} e_y\right\|^2 = \lambda_{\min}(P_2) V_s.$$

The first equality follows from $\|P_2 e_y\|^2 = \langle P_2 e_y, P_2 e_y \rangle = \langle P_2^{1/2} e_y, (P_2^{1/2})^T P_2 e_y \rangle = \langle P_2^{1/2} e_y, P_2 P_2^{1/2} e_y \rangle = (P_2^{1/2} e_y)^T P_2 (P_2^{1/2} e_y)$.

Hence in the domain

$$\Omega = \{(e_1, e_y) : \|A_{21} e_1\| < \|D_2\| \gamma_0 - \eta\}$$

where η is a small positive integer. It follows that

$$\dot{V}_s < -2\eta \|P_2 e_y\| \leq -2\eta \sqrt{\lambda_{\min}(P_2)} \sqrt{V_s}.$$

Hence output error e_y enters Ω in finite time and remains there.

Thank You!